



PRACTICAL 5

Aim: Implementation of knapsack algorithm using dynamic programming.

Algorithm:

Knapsack Algorithm

Input: Enter no. of items, Enter knapsack capacity and enter each item weight and value.

Output: Maximum value, DP Table, and selected items.

Step 1: Start

Step 2: Read n and w

Step 3: Read weights $w[i]$ and value $v[i]$ for each item.

Step 4: Initialize $dp[i][0] = 0$ and $dp[0][j] = 0$

Step 5: for $i = 1$ to n
 for $j = 1$ to w :
 if $w[i] \leq j$:
 $dp[i][j] = \max(v[i] + dp[i-1][j-w[i]], dp[i-1][j])$
 else:
 $dp[i][j] = dp[i-1][j]$

Step 6: Max^m value = $dp[n][w]$

Step 7: Set $j = w$

Step 8: for $i = n$ down to 1:
 if $dp[i][j] \neq dp[i-1][j]$:
 Item i is taken.

$j = j - w[i]$
 else:
 Item i is not taken.

Step 9: Display max^m value and selected items

Step 10: Stop.



Program / Code:

```
#include<iostream>

using namespace std;

class Knapsack {

private:

    int w[10];    // weights
    int v[10];    // values
    int n;        // number of items
    int W;        // capacity
    int dp[10][100]; // dp table

public:

    // — Get Input —
    void getInput() {
        cout << "Enter number of items  : ";
        cin >> n;
        cout << "Enter knapsack capacity : ";
        cin >> W;
        for(int i = 1; i <= n; i++) {
            cout << "Item " << i << " weight and value : ";
            cin >> w[i] >> v[i];
        }
    }
};
```



```
    }  
}  
  
// — Build dp table —  
void solve() {  
  
    // base case — row 0 and col 0  
    for(int i = 0; i <= n; i++) dp[i][0] = 0;  
    for(int j = 0; j <= W; j++) dp[0][j] = 0;  
  
    // fill table  
    for(int i = 1; i <= n; i++) {  
        for(int j = 1; j <= W; j++) {  
  
            if(w[i] <= j) {  
                // take or not take  
                dp[i][j] = max(  
                    v[i] + dp[i-1][j-w[i]],  
                    dp[i-1][j]  
                );  
            } else {  
                // cannot take item i  
                dp[i][j] = dp[i-1][j];  
            }  
        }  
    }  
}
```



```
    }  
}  
}  
  
// — Print dp table —  
void printTable() {  
    cout << "\nDP Table:" << endl;  
    cout << "    ";  
    for(int j = 0; j <= W; j++)  
        cout << j << " ";  
    cout << endl;  
  
    for(int i = 0; i <= n; i++) {  
        cout << "i=" << i << " ";  
        for(int j = 0; j <= W; j++)  
            cout << dp[i][j] << " ";  
        cout << endl;  
    }  
}  
  
// — Find selected items —  
void findItems() {  
    cout << "\nSelected Items:" << endl;  
    int j = W;
```



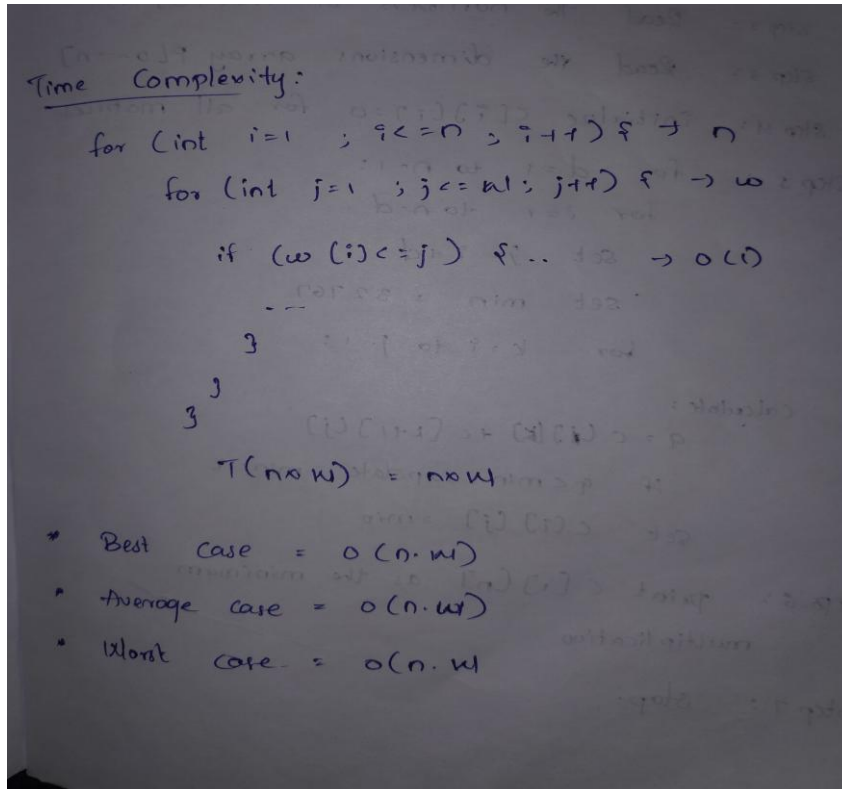
```
for(int i = n; i >= 1; i--) {  
    if(dp[i][j] != dp[i-1][j]) {  
        cout << "Item " << i  
            << " (w=" << w[i]  
            << ", v=" << v[i]  
            << ") TAKEN " << endl;  
        j = j - w[i];  
    } else {  
        cout << "Item " << i  
            << " (w=" << w[i]  
            << ", v=" << v[i]  
            << ") NOT taken " << endl;  
    }  
}  
}  
  
// — Show result —  
void showResult() {  
    cout << "\nMaximum Value = "  
        << dp[n][W] << endl;  
}  
};  
  
// — MAIN —
```



```
int main() {  
  
    Knapsack k;  
  
    k.getInput();  
    k.solve();  
    k.printTable();  
    k.showResult();  
    k.findItems();  
  
    return 0;  
}
```



Time Complexity:



Conclusion: The 0/1 Knapsack problem was successfully implemented using Dynamic Programming. The algorithm calculates the maximum possible value within the given capacity and identifies the items that should be selected.